

AI Clinical Decision Support Hackathon

General Project Documentation & Five-Day Development Plan

1. Project Overview

Project Name

AI Clinical Decision Support — Hypertension Clinical RAG

Hackathon

AI Clinical Decision Support Lite Hackathon

Project Type

Retrieval-Augmented Generation (RAG) system for evidence-grounded clinical information.

Clinical Domain

Adult Hypertension

Selected Clinical Sources

The team selected two NICE documents:

1. **Hypertension in adults: diagnosis and management (NG136)**
2. **How do I control my blood pressure? Lifestyle options and choice of medicines — Patient Decision Aid**

The NICE hypertension guideline covers adults aged 18 and over, including adults with type 2 diabetes, and focuses on accurate diagnosis, cardiovascular risk assessment, treatment, and monitoring. The guideline was originally published in 2019 and was last updated on 26 February 2026.

The patient decision aid focuses on helping patients understand lifestyle choices and medicine options for controlling high blood pressure. It explicitly presents three broad options: doing nothing, trying lifestyle changes, or taking medicines together with lifestyle changes.

2. The Problem We Are Trying to Solve

Large Language Models are very good at producing fluent answers, but fluency does not mean that the information is medically correct.

A general-purpose LLM can produce a confident answer without being able to show exactly where the information came from. In a clinical context, this is a serious problem because unsupported or outdated medical information can mislead users.

The hackathon therefore requires a system that does not simply rely on the model's internal knowledge.

Instead, our system should:

Question → Retrieve trusted evidence → Give evidence to the model → Generate a grounded answer → Show the exact source

The hackathon presentation describes the required architecture as four layers:

1. **Document Ingestion**
2. **Retrieval**
3. **Generation**
4. **Safety**

The final system must retrieve relevant evidence from official clinical documents, generate answers grounded in that evidence, provide transparent citations, and refuse questions when the available evidence is insufficient.

3. Important Safety Boundary

This project is an educational clinical decision-support prototype.

It is **NOT**:

- a doctor
- a diagnostic system
- a prescription system
- a replacement for a healthcare professional
- a system that selects individualized treatment
- a system that recommends personalized drug doses

The system must never present itself as a replacement for qualified medical judgment.

The final system should:

- provide evidence-based educational information
- show the evidence used
- clearly identify the source
- avoid unsupported medical claims
- avoid personalized diagnosis
- avoid personalized prescribing
- avoid personalized dosing
- refuse or redirect when evidence is insufficient or the request is outside the supported scope

Every layer of the architecture must preserve this boundary.

4. Why We Chose Hypertension

The hackathon allows the team to select a focused clinical topic.

The Day-1 presentation uses skin cancer / melanoma as a teaching example and provides a skin-cancer retrieval notebook as an example implementation.

That material is treated as a **teaching example**, not as the mandatory clinical scope of this project.

The actual project scope is:

Adult Hypertension

The project uses two strong and complementary NICE sources:

- one detailed clinical guideline
- one patient-oriented decision aid

This combination gives the knowledge base both:

- formal clinical management evidence
- patient-friendly decision-support information

This distinction must remain explicit throughout the documentation:

Day-1 Presentation Teaching Example

↓

Skin Cancer / Melanoma

Actual Project Implementation

↓

Adult Hypertension

↓

NICE NG136

+

NICE Patient Decision Aid

5. Hypertension Knowledge Base

5.1 Diagnosis

The NICE guideline covers:

- measuring blood pressure
- measuring both arms
- repeated clinic measurements
- ambulatory blood pressure monitoring (ABPM)
- home blood pressure monitoring (HBPM)
- confirming hypertension
- cardiovascular risk assessment
- target-organ damage assessment

For example, the guideline addresses the use of ABPM to confirm hypertension when clinic blood pressure is in the relevant range, with HBPM used when ABPM is unsuitable or not tolerated.

The system should retrieve the relevant source evidence rather than relying on model memory.

5.2 Cardiovascular Risk and Target Organ Damage

The guideline covers formal cardiovascular risk estimation and investigations such as:

- urine albumin:creatinine ratio
- haematuria testing
- HbA1c
- electrolytes
- creatinine
- estimated glomerular filtration rate
- total cholesterol
- HDL cholesterol
- retinal examination
- 12-lead ECG

These are examples of clinical information available within the approved knowledge base.

5.3 Lifestyle Management

The patient decision aid presents lifestyle options including:

- healthy diet
- reducing salt
- eating more fruit and vegetables
- physical activity
- achieving a healthier weight
- avoiding excessive alcohol
- stopping smoking

It also explains advantages and disadvantages of lifestyle changes compared with medication.

5.4 Medication Information

The patient decision aid describes major medication categories including:

- ACE inhibitors
- ARBs
- calcium channel blockers (CCBs)
- diuretics

It also explains common side effects and monitoring requirements.

The documents include additional contextual information such as:

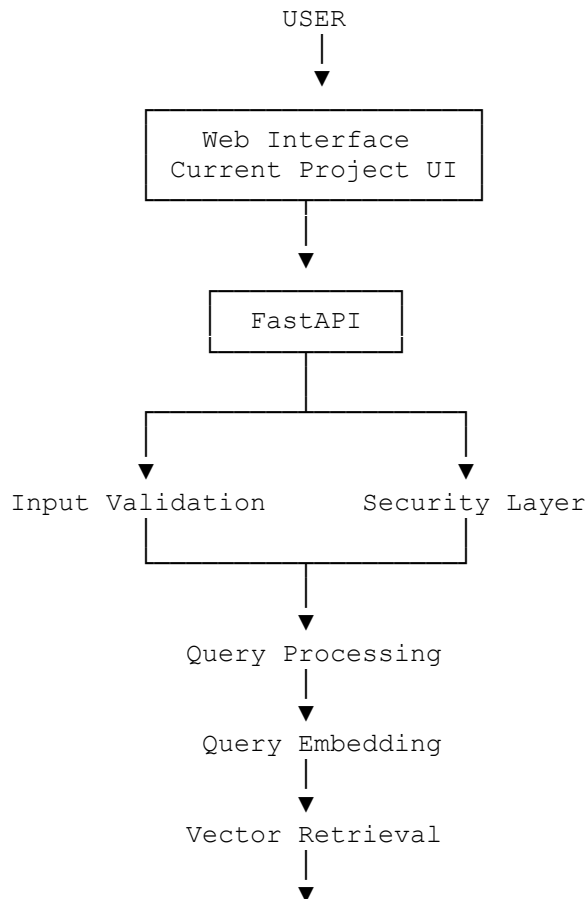
- pregnancy
- diabetes
- family origin
- kidney monitoring
- medicine-specific considerations

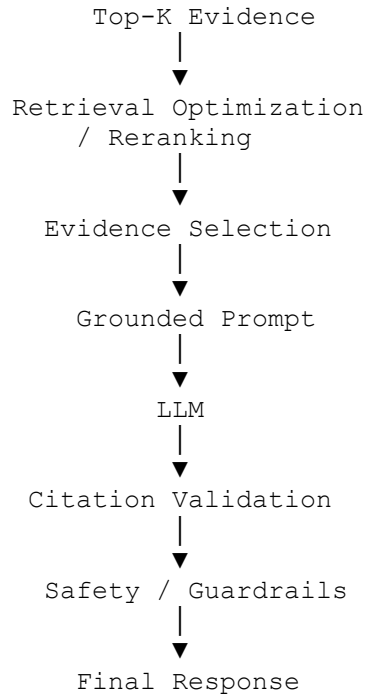
The system must present this information as evidence-grounded educational content and must not convert it into individualized prescribing advice.

6. System Architecture

The project follows a modular RAG architecture.

The complete target architecture is:





The architecture follows the four required layers:

Layer 1 → Document Ingestion
Layer 2 → Retrieval
Layer 3 → Generation
Layer 4 → Safety

7. Canonical Frontend Architecture

Important Architecture Decision

The original planning documentation recommended:

Next.js + React + TypeScript

That recommendation is no longer treated as a mandatory implementation requirement.

The project already has a working Day-1 web interface, and the project decision is to **keep the current Day-1 UI architecture and improve it incrementally** rather than replacing it with a new frontend framework.

Therefore, the current UI is now the:

Canonical Project UI

The final architecture is:

```
Current Day-1 Web UI
      ↓
FastAPI Backend
      ↓
RAG Pipeline
```

The frontend technology itself is not the core objective of the hackathon.

The important requirements are that the UI should clearly expose:

- question
- retrieved evidence
- similarity score
- rank
- document
- section
- page
- chunk ID
- final grounded answer
- citations
- safety behavior

The existing UI should therefore be preserved and improved rather than replaced solely to adopt a different frontend framework.

8. Current Web UI Direction

The Day-1 UI is intentionally lightweight.

Its purpose is to demonstrate:

```
Question
      ↓
Search
      ↓
Retrieved Evidence
      ↓
Metadata
```

The Evidence Panel should make the retrieval process transparent.

Minimum displayed evidence metadata:

- Rank
- Similarity
- Document
- Section
- Page
- Chunk ID
- Evidence text

As the project evolves, the same UI can be extended to display:

- grounded answer
- citation information
- evidence strength
- safety status
- refusal / redirect messages
- evaluation information

The UI should evolve from the current Day-1 foundation instead of being replaced.

9. Backend Architecture

Backend

FastAPI + Python

Reasons:

- the RAG pipeline is already implemented in Python
- easy integration with Chroma
- easy integration with embedding libraries
- API documentation support
- clean separation between UI and RAG logic
- request validation support

The backend is responsible for:

- receiving requests
- validating input
- processing queries
- retrieving evidence
- passing evidence to the generation layer
- returning structured results
- exposing retrieval metadata

10. Knowledge Base Architecture

The ingestion pipeline is:

```
Approved NICE PDFs
  ↓
PDF Parsing
  ↓
Text Cleaning
  ↓
Section Detection
  ↓
Section-Aware Chunking
  ↓
Embeddings
  ↓
Chroma Vector Database
```

The vector database stores:

```
Chunk Text
+
Embedding
+
Metadata
```

The metadata connects every retrieved vector back to the original clinical source.

Minimum metadata:

```
document_name
section
page_number
```

Recommended metadata:

```
document_id
source
version
source_url
chunk_id
```

11. Metadata and Traceability

Every chunk should remain traceable to its source.

Example:

```
{
  "document_name": "Hypertension in adults: diagnosis and management",
  "section": "1.2 Diagnosing hypertension",
  "page_number": 7,
  "chunk_id": "HTN-NG136-P007-C001"
}
```

The metadata design is particularly important because the final system must be able to explain:

Where did this evidence come from?

The answer should be:

```
Document
→ Section
→ Page
→ Chunk ID
```

12. Security Plan

Security should not be treated as something added at the end.

API Security

The project should include:

- environment variables for secrets
 - no API keys inside source code
 - CORS restrictions
 - request validation
 - rate limiting
 - maximum input length
 - structured API errors
-

RAG Security

The retrieval system should:

- restrict retrieval to approved documents
 - prevent arbitrary external URLs from becoming knowledge sources
 - validate document metadata
 - prevent unauthorized modification of the vector database
 - maintain document/version information
-

Prompt Injection Protection

Retrieved medical text should be treated as evidence, not executable instructions.

The system should not blindly follow instructions contained inside retrieved documents.

Privacy

The application should avoid storing personal medical information.

The demo should use synthetic or non-identifying questions.

Logging

Where appropriate, the system may log:

- query ID
- timestamp
- retrieval scores
- retrieved chunk IDs
- model response status
- refusal status

The system should not log unnecessary personal information.

13. Hackathon Evaluation Criteria

The project evaluation is:

Criterion	Weight
Retrieval Precision	30
Grounding & Citations	25
Architecture Design	15
Evaluation Metrics	15
Clinical Safety	10
UX & Live Demo	5
Total	100

These weights should guide development priorities.

Recommended priority:

1. Retrieval quality
2. Grounding and citations
3. Evaluation
4. Architecture
5. Safety
6. UX polish

Safety remains mandatory regardless of its relative score.

14. Five-Day Development Plan

Day 1 — Research, Scope & Ingestion

Main Focus

Research, clinical scope, source selection, PDF ingestion, chunking, embeddings, and vector indexing.

Main Output

Searchable Vector Database

Tasks:

- select 1–2 official guideline PDFs
 - verify public accessibility
 - review legal usability
 - parse PDFs
 - implement section-aware chunking
 - generate embeddings
 - index chunks
 - store document name, section, and page metadata
 - implement baseline retrieval
 - create Day-1 test questions
 - demonstrate retrieved evidence through the UI
-

15. Day 1 Team Allocation

Member 1 — Clinical Research & Scope

Responsibilities:

- verify NICE sources
 - document clinical scope
 - review licensing/reuse requirements
 - create source documentation
 - identify important clinical sections
 - prepare clinical test questions
-

Member 2 — PDF Ingestion

Responsibilities:

- implement PDF parsing
- preserve page numbers
- clean extraction artifacts
- validate extracted text

Member 3 — Chunking & Metadata

Responsibilities:

- implement section-aware chunking
- test chunk size
- preserve section/page
- generate stable chunk IDs
- validate metadata

Member 4 — Embeddings & Vector Database

Responsibilities:

- select embedding model
- configure Chroma
- configure cosine similarity
- implement indexing
- implement retrieval API
- verify vector database persistence

Member 5 — QA + UI/Security Foundation

Responsibilities:

- create Day-1 test questions
 - validate retrieval output
 - identify bad chunks
 - maintain the basic web UI
 - configure environment variables
 - establish security baseline
 - perform integration testing
-

16. Day 2 — Retrieval Optimization

Day 2 is dedicated to improving the retrieval baseline.

Chunk Size Experiments

Compare:

- 400–600 tokens
- 700–900 tokens

The current Day-1 implementation uses:

```
850 characters  
150 characters overlap
```

This is treated as the **Day-1 baseline**.

Characters and tokens are not equivalent, so the Day-2 experiments should move toward token-based chunking.

Top-K Experiments

Compare:

- Top-3
 - Top-5
 - Top-10
-

Retrieval Strategies

Compare:

- semantic/vector search
 - keyword/BM25
 - hybrid search
 - reranking
-

Evaluation Dataset

Build approximately:

15–20 labeled clinical questions

Measure:

- Precision@3
- Precision@5
- retrieval score
- relevant evidence rate

The current Day-1 set of 8 questions is considered a baseline and should be expanded during Day 2.

17. Evidence Panel

The Evidence Panel should show:

- question
- retrieved chunks
- similarity score
- rank
- page
- section
- document
- chunk ID

The current Day-1 UI should remain the foundation for this panel.

The UI should become more polished without changing the underlying interaction model.

18. Day 3 — Grounded Generation & Citations

The retrieved evidence becomes the input context for the LLM.

The model must behave as an:

Evidence Synthesizer

rather than an independent medical authority.

Expected response structure:

Answer

Key Points:

- ...
- ...
- ...

Evidence:

[Document Name]

Section: ...

Page: ...

Chunk ID: ...

Evidence Strength:

High / Medium / Low / Insufficient

Educational Disclaimer

The citation should identify:

- document
 - section
 - page
 - chunk ID
-

19. Day 4 — Safety, Guardrails & Evaluation

The agenda defines three major safety mechanisms.

19.1 Input Risk Classification

Classify queries as:

- Allowed
 - Needs Caution
 - Refuse / Redirect
-

19.2 Retrieval Confidence Threshold

If retrieval quality is too low:

`Do not generate a confident answer.`

Instead:

`Insufficient evidence.`

`Please consult a qualified healthcare professional.`

19.3 Unsupported Claim Detection

After generation:

1. extract claims
2. compare claims against retrieved evidence
3. detect unsupported claims
4. reject or downgrade unsafe responses

RAG does not eliminate all risk.

Potential failure points include:

- PDF parsing errors
- incorrect section detection

- poor chunking
 - lost context
 - incorrect retrieval
 - LLM misinterpretation of evidence
-

20. Day 5 — Final Presentation & Judge Evaluation

The prototype should be frozen before the final presentation.

The demo should include three predefined scenarios.

Scenario 1 — Simple Query

A question answered from a single relevant chunk.

Scenario 2 — Multi-Chunk Query

A question requiring evidence from multiple sections.

Scenario 3 — Unsafe / Unsupported Query

A question outside the knowledge base.

Expected behavior:

The system does not invent an answer.

It explains that the available evidence is insufficient and redirects the user appropriately.

21. Final Demo Must Show

The final demonstration should show:

- web interface
 - question
 - retrieved evidence
 - source
 - page
 - section
 - similarity score
 - final grounded answer
 - citation
 - safety behavior
 - evaluation metrics
 - architecture
-

22. Definition of Done

The complete system should eventually demonstrate:

```
User Question
      ↓
Input Validation
      ↓
Safety Classification
      ↓
Query Embedding
      ↓
Vector Search
      ↓
Top-K Retrieval
      ↓
Reranking
      ↓
Evidence Selection
      ↓
Grounded LLM Prompt
      ↓
Generated Answer
      ↓
Citation Verification
      ↓
Safety Verification
      ↓
Final Answer
```

The final answer must remain traceable to the approved clinical documents.

23. Final Team Principle

The main idea of the project is not:

"Let's build a chatbot that knows medicine."

The real idea is:

"Let's build a system that knows where its medical evidence comes from, retrieves that evidence correctly, shows it transparently, and refuses to invent information when the evidence is insufficient."

That is the mindset the entire team should follow throughout the five days.